



# The Volkswagen emissions scandal: Exploring the role of environmental concern and social norms<sup>☆</sup>

M. Marino<sup>a</sup>, P. Parrotta<sup>b,c,d,\*</sup>, D. Sala<sup>e</sup>, G. Valletta<sup>f</sup>

<sup>a</sup> SKEMA Business School - Université Côte d'Azur, Avenue Willy Brandt, 59777 Euralille, France

<sup>b</sup> University of Siena, Piazza San Francesco 7-8, 53100 Siena, Italy

<sup>c</sup> IZA - Institute of Labor Economics, Schaumburg-Lippe-Straße 5-9, 53113 Bonn, Germany

<sup>d</sup> NoCeT - Norwegian Centre for Taxation, Helleveien 30, 5045 Bergen, Norway

<sup>e</sup> University of Passau, 94030 Passau, Germany

<sup>f</sup> EDHEC Business School, 24 avenue Gustave Delory, 59057 Roubaix, France

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## ABSTRACT

Nearly a decade after the 2015 emissions scandal, Volkswagen has transitioned from marketing deceptive 'clean engine cars' to becoming a major player in the electrification of the automotive industry. Yet, the violation of environmental standards during the scandal resulted in excessive pollutant emissions, posing persistent threats to health and the environment. This paper explores how consumers held Volkswagen accountable for these hazards. Our analysis reveals that the decline in Volkswagen's sales volumes following the scandal was driven by *environmentally concerned* consumers globally. However, their reaction was short-lived and mainly limited to the models implicated by the U.S. Environmental Protection Agency. Interestingly, we find no evidence of lost sales to competitors. Instead, we document a *within-brand* or *within-group* shift in favor of Volkswagen models compliant with environmental regulations. This phenomenon could be one of the contributing factors explaining Volkswagen's resilience in navigating the fallout from the scandal.

## 1. Introduction

The Volkswagen emissions scandal will certainly be remembered as one of the largest and most prominent scandals in the history of the automotive sector, not just for being a setback in the image of the 80-year-old German industry leader, but also for its far-reaching environmental implications, with recognized repercussions also on human health (e.g., Chossière et al., 2017; Alexander and Schwandt, 2022).

The scandal erupted in September 2015, when the US Environmental Protection Agency (EPA henceforth) accused the Volkswagen Group of having manipulated some of its diesel models to deceive emissions tests.<sup>1</sup> Fast forwarding to almost ten

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\* Corresponding author at: University of Siena, Piazza San Francesco 7-8, 53100 Siena, Italy.

E-mail addresses: [marianna.marino@skema.edu](mailto:marianna.marino@skema.edu) (M. Marino), [pierpaolo.parrotta@unisi.it](mailto:pierpaolo.parrotta@unisi.it) (P. Parrotta), [davide.sala@uni-passau.de](mailto:davide.sala@uni-passau.de) (D. Sala), [giacomo.valletta@edhec.edu](mailto:giacomo.valletta@edhec.edu) (G. Valletta).

<sup>1</sup> At the time of the emissions scandal, Volkswagen AG, known as Volkswagen Group, included the following passenger car brands: Audi, Bentley, Bugatti, Lamborghini, Porsche, SEAT, Škoda, Volkswagen.

years from the outbreak of the scandal, Volkswagen has not only survived the significant legal fines and recall costs, but it has also managed to drastically remodel itself, from “Emissions Cheater to Climate Leader”, to have a leading role in the electrification of the automotive industry.<sup>2</sup>

However, to what extent have Volkswagen’s customers held the company accountable for its malfeasance? In particular, did their environmental concern and their culture play a role in their reaction? Can these characteristics be associated with different consumer responses across different markets? Surprisingly, these aspects of the scandal have so far received almost no attention. Shedding light on these questions will enhance our understanding of the multifaceted implications of the scandal, which is the aim of our study.

To address our questions, we rely on a multi-market approach: this is made possible by the fact that we can exploit an international data set on new car registrations. The international scope of the dataset allows us to explore the relationship between consumer responses to the scandal across different countries and their level of environmental concern and their attitude towards social norms. For both these traits we rely on pre-existing measures at the country level.<sup>3</sup> Consumers concerned about environmental decay tend to support stricter environmental regulations (Kahn, 2007; Schumacher, 2014), to pay a premium for environmentally friendly goods (Martinez, 2011) and to have a higher awareness of environmental issues (Schlegelmilch et al., 1996) and challenges (Laroche et al., 2001). These consumers are hence more likely to hold Volkswagen accountable for a misconduct that is harmful to the environment. Likewise, consumers from countries with lower tolerance for deviations from social norms tend to respond more severely to corporate misconduct (Li et al., 2017; Torelli and Rodas, 2017).

In line with previous contributions, we rely on the fact that the scandal was largely an unexpected event and, using it as a source of exogenous variation, we deploy a difference-in-difference (DiD henceforth) estimation strategy. The granularity of our data (at the model-maker level) allows us (a) to include a wide set of fixed effects to eliminate a number of relevant unobserved confounding factors and (b) to perform our analysis (and present our results) by car segments.<sup>4</sup> This allows us to gauge the complexities involved in evaluating consumers’ response to the scandal, revealing to some extent their propensity to substitute away from the Volkswagen brand.

Our analysis reveals that social norms, and primarily consumers’ tolerance for deviations from them, have hardly any explicative power regarding Volkswagen’s foregone sales due to the scandal. On the contrary, the punishment triggered by *environmentally concerned* consumers turns to be significant even if short-lived. Within a year from the outbreak of the scandal, the decline in Volkswagen’s sales volumes in the market with the highest level of environmental concern is between 50 and 60 percentage points higher than in the market with the lowest level. The punishment ascribable to environmentally concerned consumers regards essentially the models initially investigated by EPA, most of which belonged to the segments C and D, namely medium and large cars in the Euro Car Segments classification.<sup>5</sup> The Volkswagen Group models incriminated by the EPA within these segments include Audi A3, Beetle, Beetle Convertible, Golf, Golf Sportwagen, Jetta, Jetta Sportwagen, and Passat.<sup>6</sup> Interestingly, the segment-by-segment analysis reveals that substitution away from the models directly implicated in the scandal, in segments C and D, has not necessarily favored competitors’ models. Indeed, substitution occurred also within the Volkswagen brand but between different car segments (i.e., involving models of different size, for example, a “Polo” for a “Golf”) or within the Volkswagen Group (i.e., involving models belonging to different brands of the Group, for example, a “Skoda Octavia” instead of a “Passat”). These substitution patterns may help explaining the group’s resilience at weathering the storm: they seem to suggest that, to some extent, consumers switched from scandal-affected models to other Volkswagen Group models rather than to models produced by competitors.

The empirical landscape concerning the extent and nature of consumer responses to the scandal, especially in terms of foregone sales of new cars, remains quite scarce. Existing studies on the scandal typically utilize data about second-hand cars, rather than new car registrations, and focus on a single country or market, thereby rendering it impossible to accurately assess the extent of consumer responses to the scandal across national markets. Ater and Yoseph (2022), Che et al. (2020), and Strittmatter and Lechner (2020) all find that consumers have to some degree penalized the Volkswagen brand in the used car market, respectively in Israel, USA, and Germany. Bachmann et al. (2023) rely on data on new car registrations in the US. However, they focus on the reputational spill-over, due to the scandal, on other German brands: they find that both the sales of Daimler and BMW have decreased after the scandal. Furthermore, other papers exploring the scandal have concentrated on its impact within financial markets, as demonstrated by Jourdan et al. (2019) and Barth et al. (2022). Thus, to the best of our knowledge, a broad analysis of the potential heterogeneity of consumer reactions to the scandal, across countries, is still missing in the literature.

The paper is organized as follows: Section 2 sketches the unfolding of the scandal. Section 3 describes our data and the estimation strategy. Section 4 presents our findings about segments C and D, to which most of the investigated models belonged to. We first

<sup>2</sup> See the article by Georg Kell “From Emissions Cheater To Climate Leader: VW’s Journey From Dieselgate To Embracing E-Mobility” (*Forbes* - December 5th 2022).

<sup>3</sup> We use an index of environmental concern derived from the share of votes garnered nationally by political parties advocating climate change mitigation or environmental protection, which we retrieved from the Manifesto Research Database. To measure the attitude towards social norms we use the looseness-tightness index developed by Gelfand (2011, 2018).

<sup>4</sup> In this paper, we adopt the car classification provided by our data source, MarkLines Automotive Industry Portal. This classification aligns with the Euro Car Segments classification, which is widely used in the European Union (European Commission, 1999). Our classification encompasses the following segments: A (mini-cars), B (small cars), C (medium cars), D (large cars), E (executive cars), F (luxury cars), MPV (multi-purpose vehicles), and SUV (sport utility vehicles). All vehicles in this classification are (passenger) cars, so the words “car” and “vehicle” are used interchangeably throughout the paper. For a detailed description of these segments, including examples of the car types they encompass, please refer to Table A.1.

<sup>5</sup> These two segments correspond to the Compact and Mid-size segments commonly used in the US classification (see Table A.1).

<sup>6</sup> For an overview of other models that were implicated by the EPA at that time and additional models produced by the Group, please refer to Table A.3.

**Table 1**  
Sample composition by year.

| Year | Months  | Segments C & D |        | Segments A & B |        | Higher segments |        | All segments |        |
|------|---------|----------------|--------|----------------|--------|-----------------|--------|--------------|--------|
|      |         | Obs            | %      | Obs            | %      | Obs             | %      | Obs          | %      |
| 2014 | Sep–Dec | 4,743          | 15.12  | 2,582          | 14.78  | 7,298           | 13.77  | 14,623       | 14.36  |
| 2015 | Jan–Dec | 15,290         | 48.76  | 8,555          | 48.96  | 26,017          | 49.07  | 49,862       | 48.96  |
| 2016 | Jan–Sep | 11,326         | 36.12  | 6,338          | 36.27  | 19,703          | 37.16  | 37,367       | 36.69  |
| All  |         | 31,359         | 100.00 | 17,475         | 100.00 | 53,018          | 100.00 | 101,852      | 100.00 |

Note: Car segments follow the European passenger car classification: Segments A and B (*mini and small cars*), Segments C and D (*medium and large cars*); *Higher Segments* is an aggregate of segment classes (segments E, F, MPV, and SUV).

provide results from the pooled sample and then discuss our main findings by analyzing consumer reactions across different markets, associating them with their level of environmental concern and their attitude towards social norms. Section 5 explores our findings about potential substitution patterns within the Volkswagen (VW henceforth) brand or Volkswagen Group (VWG henceforth), as well as across automotive groups. Section 6 concludes.

## 2. The unfolding of the scandal

On September 18th, 2015, the EPA announced that the German automaker Volkswagen had allegedly installed an illegal software, on some of its diesel models, that was able to manipulate, during a test, the amount of emissions produced. According to the EPA, when the defeat device was turned off (i.e., in normal driving conditions), the vehicle could release between 10 and 40 times as much of the pollutant nitrogen oxide that was allowed by the Clean Air Act in USA. Among other pollutants, nitrogen oxide has been linked to a number of health issues, including respiratory problems.

A few days after the EPA announcement the company admitted that the defeat device was installed in many cars of the group, produced between 2009 and 2014 and equipped with a TDI diesel engine, mainly with a cylinder capacity of 1.6 L and 2.0 L. Ironically, this was the engine that had been central in the “clean diesel” marketing campaign implemented globally. Since the industry was (already) under a growing pressure for a re-evaluation of the sustainability of their products and value chains, VWG decided to bet on a new clean diesel engine instead of investing in electric or hybrid cars. This explains why, as largely documented, the scandal came as a total surprise for the public and the media, which we exploit to our great advantage in our empirical strategy.<sup>7</sup>

After the news became public, VW's share plunged and some observers were even wondering if the German company would “ever recover” while, in a few countries, the sales of the models implicated in the scandal were temporarily halted.<sup>8</sup>

So far, the scandal has already costed the company more than \$35 billion in fines, penalties, recalls and restitutions, legal fees and settlements while, in several countries, investigations and litigation are still pending. In spite of all this, the VWG managed to bounce back in a rather surprising fashion. Beside restructuring the organization (see Welch (2019), Jung and Sharon (2019)) it has greatly focused on electrification, connectivity and mobility: not only to follow the main trends in the industry, but also to move as fast as possible away from the image created by the emissions scandal.

Of course, the prompt recovery in sales does not reveal whether and to what extent sales volumes of models featuring the defeat device were directly impacted by the scandal (compared to other models within and outside the VWG) and whether this response has been heterogeneous across countries.

## 3. Data and methodology

### 3.1. Data

Our empirical analysis uses data collected from the MarkLines Automotive Industry Portal, providing monthly records on the number of newly-registered cars over a large number of countries, for the period 2012–2017.<sup>9</sup> The information on each car is equally rich, comprising the model, the maker, and the country of sale, and most importantly, it is consistent and highly comparable across markets. As mentioned already, the data follows the European passenger car classification described in Table A.1. Throughout our analysis we use three segment aggregates: segments A and B, segments C and D, and *higher segments*, including all segments above segments C and D.

Table 1 describes the sample composition by year, whereas Table 2 illustrates its composition by country for the sample period. As shown in the Appendix (cf. Table A.2), our sample effectively reflects the relative market share of each industrial group on a global scale. As it will be clearer below, the bulk of our analysis rests on segments C and D. Over the shorter time span in between

<sup>7</sup> For example, Bachmann et al. (2023) show that the monthly print media mentions of the word “Volkswagen” spiked concurrently with the EPA announcement, suggesting that this event was absolutely unexpected. Similarly, as showed by Barth et al. (2022), before VW's admission of wrongdoing, there has not been any significant reaction in VW's stocks or bonds, consistently with the fact that there were no pre-event leakages.

<sup>8</sup> See <https://europe.autonews.com/article/20150929/ANE/150929810/volkswagen-plans-to-refit-diesel-cars-affected-by-emissions-scandal> and <https://www.bbc.com/news/business-34410390>.

<sup>9</sup> This is a subscription-based database, subject to all the terms and conditions stated on MarkLines Co., Ltd.'s website <https://www.marklines.com>.

**Table 2**  
Sample composition by country (September 2014–September 2016).

| Country        | Segments C & D |        | Segments A & B |        | Higher segments |        | All segments |        |
|----------------|----------------|--------|----------------|--------|-----------------|--------|--------------|--------|
|                | Obs            | %      | Obs            | %      | Obs             | %      | Obs          | %      |
| Australia      | 2,149          | 6.85   | 659            | 3.77   | 3,208           | 6.05   | 6,016        | 5.91   |
| Austria        | 1,464          | 4.67   | 970            | 5.55   | 2,594           | 4.88   | 5,023        | 4.93   |
| Canada         | 1,858          | 6.05   | 417            | 2.39   | 3,520           | 6.64   | 5,835        | 5.73   |
| Finland        | 1,255          | 4.00   | 782            | 4.47   | 2,098           | 3.96   | 4,135        | 4.06   |
| Germany        | 1,858          | 5.92   | 1,118          | 6.40   | 3,498           | 6.60   | 6,474        | 6.36   |
| Greece         | 1,003          | 3.20   | 996            | 5.70   | 1,136           | 2.14   | 3,135        | 3.08   |
| Ireland        | 1,345          | 4.29   | 882            | 5.05   | 2,046           | 3.38   | 4,273        | 4.20   |
| Israel         | 1,807          | 5.76   | 1,025          | 5.87   | 2,629           | 4.96   | 5,461        | 5.36   |
| Italy          | 1,567          | 5.00   | 1,231          | 7.04   | 3,038           | 5.73   | 5,836        | 5.73   |
| Mexico         | 1,608          | 5.13   | 558            | 3.19   | 2,102           | 3.96   | 4,268        | 4.19   |
| Netherlands    | 1,730          | 5.52   | 1,116          | 6.39   | 3,404           | 6.42   | 6,250        | 6.14   |
| Poland         | 1,920          | 6.12   | 1,148          | 6.57   | 3,680           | 6.94   | 6,748        | 6.63   |
| Portugal       | 1,575          | 5.02   | 1,147          | 6.56   | 2,515           | 4.74   | 5,237        | 5.14   |
| Slovakia       | 1,538          | 4.90   | 900            | 5.15   | 2,691           | 5.08   | 5,129        | 5.04   |
| Spain          | 1,769          | 5.64   | 1,071          | 6.13   | 3,174           | 5.99   | 6,014        | 5.90   |
| Sweden         | 1,405          | 4.48   | 835            | 4.78   | 2,461           | 4.64   | 4,701        | 4.62   |
| Turkey         | 1,674          | 5.34   | 904            | 5.17   | 2,850           | 5.38   | 5,428        | 5.33   |
| United Kingdom | 1,696          | 5.41   | 1,199          | 6.86   | 2,804           | 5.29   | 5,699        | 5.60   |
| United States  | 2,098          | 6.69   | 517            | 2.96   | 3,575           | 6.74   | 6,190        | 6.08   |
| All            | 31,359         | 100.00 | 17,475         | 100.00 | 53,018          | 100.00 | 101,852      | 100.00 |

Note: Car segments follow the European passenger car classification: Segments A and B (*mini and small cars*), Segments C and D (*medium and large cars*); *Higher Segments* is an aggregate of segment classes (segments E, F, MPV, and SUV).

**Table 3**  
Environmental concern and cultural tightness indexes (normalized values).

| Country        | Environmental Concern (MRG) | Country        | Environmental Concern (Franzen and Vogl, 2013) | Country        | Cultural Tightness (Eriksson et al., 2021) |
|----------------|-----------------------------|----------------|------------------------------------------------|----------------|--------------------------------------------|
| Canada         | 1.00                        | Canada         | 1.00                                           | Sweden         | 1.00                                       |
| United Kingdom | 0.99                        | Finland        | 0.86                                           | Turkey         | 0.95                                       |
| Greece         | 0.81                        | Sweden         | 0.80                                           | Austria        | 0.84                                       |
| Mexico         | 0.70                        | Austria        | 0.54                                           | Germany        | 0.72                                       |
| Sweden         | 0.59                        | Australia      | 0.53                                           | Portugal       | 0.67                                       |
| Austria        | 0.54                        | Spain          | 0.51                                           | Australia      | 0.50                                       |
| United States  | 0.45                        | United States  | 0.50                                           | Italy          | 0.46                                       |
| Italy          | 0.43                        | Israel         | 0.27                                           | Canada         | 0.39                                       |
| Germany        | 0.40                        | United Kingdom | 0.20                                           | United States  | 0.38                                       |
| Poland         | 0.29                        | Mexico         | 0.19                                           | Ireland        | 0.34                                       |
| Portugal       | 0.27                        | Slovakia       | 0.11                                           | United Kingdom | 0.29                                       |
| Finland        | 0.26                        | Turkey         | 0.00                                           | Finland        | 0.26                                       |
| Ireland        | 0.21                        |                |                                                | Greece         | 0.20                                       |
| Slovakia       | 0.20                        |                |                                                | Spain          | 0.20                                       |
| Netherlands    | 0.15                        |                |                                                | Slovakia       | 0.19                                       |
| Spain          | 0.14                        |                |                                                | Poland         | 0.18                                       |
| Australia      | 0.12                        |                |                                                | Mexico         | 0.16                                       |
| Israel         | 0.01                        |                |                                                | Israel         | 0.11                                       |
| Turkey         | 0.00                        |                |                                                | Netherlands    | 0.00                                       |

Note: The *environmental concern index*, in the left panel, quantifies citizens' support in favor of "environmental protection" using a 0–1 scale (value "1" expressing the highest concern). It is derived from the *Manifesto Research Database* based on the collective share of votes obtained (in national elections next closest in time to the pre-sample year) by parties whose political Manifesto endorse the fight to climate change or environmental preservation policies. The *cultural tightness index* from Eriksson et al. (2021), in the right panel, reflects the degree of *tolerance* towards *noncompliance* with *social norms*. A higher (normalized) value along the 0–1 scale indicates a higher degree of the *pervasiveness of social norms*. The index of *environmental concern*, in the middle panel, (available for a subset of countries in our sample) is the one popularized by Franzen and Vogl (2013). Countries are listed in descending order of each respective index.

September 2014 and September 2016, just at the straddle of the scandal, we have a total of 101,852 observations, 31,359 belong to segments C and D.

The sample encompasses 19 countries, including most of the prominent national markets in the automotive sector. Such richness in terms of international coverage comes at a price: we miss the information regarding the type of engine mounted on the sold car. As we clarify below, this does not represent such a limiting factor for our intent as it may at first seem, since – if anything – it leads to an underestimation of the consequences of VW emissions scandal on the company's sales.

The multi-country dimension of our data is the feature that sets our analysis apart from other studies as it opens to the exploration of market idiosyncrasies in consumer reactions to the scandal worldwide. Our primary aim is establishing whether this heterogeneity is associated with some consumers' traits, like their attitude towards the environment or social norms.

To measure *environmental concern* at the country level, we use an index constructed from the Manifesto Research Database, that is based on the share of votes secured in national elections (the closest in time before the scandal) by political parties whose manifestos advocate climate change mitigation or, more broadly, environmental preservation.<sup>10</sup> To corroborate our findings, we also use the environmental concern index proposed by [Franzen and Vogl \(2013\)](#) for a robustness check (this index is only available for a subset of countries within our data set).<sup>11</sup>

To measure the pervasiveness of social norms across the countries in our study, we rely on the *cultural tightness* index proposed by [Gelfand et al. \(2011\)](#): tight cultures exhibit more rigid norms and a lower tolerance for deviations from them ([Gelfand, 2018](#)). Our measure is drawn from [Eriksson et al. \(2021\)](#), that uses the same index proposed in [Gelfand et al. \(2011\)](#), extended to a broader set of countries, and it reflects both the strength of social norms within a country and the extent of tolerance for those who violate them.<sup>12</sup> A higher score corresponds to a higher degree of cultural tightness.

To ease the interpretation of our results, we have re-scaled our measures of environmental concern and of cultural tightness so that they fall within the unit interval (with higher values denoting a greater levels of environmental concern or cultural tightness). [Table 3](#) displays the re-scaled values of each of the two indexes.

### 3.2. Empirical strategy

The unfolding of VW emissions scandal makes an empirical strategy centered on the DiD methodology particularly well suited. As discussed above, the revelation of the scandal to the general public was sudden and unexpected, and caught most consumers worldwide off guard, making expectation-driven reactions on their side very improbable.<sup>13</sup>

As we observe car registrations on a monthly basis, it is convenient to introduce a notation that distinguishes between the calendar time,  $t$ , expressing a specific *month-year* combination, and just a particular month of the year, indexed hereafter with  $\kappa$ . In keeping with the quasi-experimental setting offered by the VW emissions scandal, we purposely restrict our analysis to a quite narrow time window, between September 2014 and September 2016, just straddling the revelation of the scandal in September 2015. Our baseline DiD-specification reads as follows:

$$\Delta_{\kappa} \ln Sales_{mct} = \alpha + \delta \cdot Post_t + \beta^g \cdot [T_m^g \times Post_t] + f_{c \times \kappa} + f_{c \times m} + f_{\kappa \times m} + \epsilon_{mct}. \quad (1)$$

The variable  $Sales_{mct}$  is the number of new cars, whose model is  $m$ , registered in country  $c$  at the calendar time  $t$ . Given our focus on sales differentials and, at the same time, to neutralize the known seasonality of car sales, our dependent variable is expressed as the change in the logarithm of newly-registered cars with respect to the same month ( $\kappa$ ) a year earlier.<sup>14</sup>  $Post$  is the indicator function for our post-treatment period, taking value 1 from September 2015 onward. The car model (and not the maker or the brand) is the unit of our analysis throughout the paper, but the specific definition of treated models,  $T_m^g$ , is not univocal in our setting. As explained below, it will be indeed purposely varied to appreciate different nuances of the economic impact of the scandal on corporate sales. The super-script  $g$ , therefore, simply indexes one of the three different treatment groups introduced below.

*Country-month* ( $f_{c \times \kappa}$ ), *country-model* ( $f_{c \times m}$ ), and *month-model* ( $f_{\kappa \times m}$ ) fixed effects complete our specification and absorb in themselves also the singular treatment dummy typical of a DiD specification. Their function is primarily to control for a whole host of confounding factors, which are unobservable in our data, yet impact sales volumes.<sup>15</sup>

The parameter of major interest is  $\beta^g$ , our treatment effect, which represents the difference in sales performance, measured in percentage points, between the treated models and the models in the control group, after the scandal compared to before the scandal.

Moreover, to assess the relationship between consumers' reactions to the scandal and their level of environmental concern (*EnvConc*) and cultural tightness (*CulTight*) respectively, we *augment* the baseline specification, described by Eq. (1), to account for these economic channels as follows:

$$\Delta_{\kappa} \ln Sales_{mct} = \alpha + \delta \cdot Post_t + \beta^g \cdot [T_m^g \times Post_t] + \underbrace{\gamma^g \cdot [T_m^g \times Post_t \times EnvConc_c]}_{\text{Environmental Channel}} + \underbrace{\xi^g \cdot [T_m^g \times Post_t \times CulTight_c]}_{\text{Cultural Channel}} +$$

<sup>10</sup> These policies correspond to those categorized as *per501* in the code-book of the *Manifesto Project Database*, falling under the "Environmental Protection" category.

<sup>11</sup> The correlation between these two indexes is statistically significant and roughly equal to 0.36.

<sup>12</sup> The index is constructed by averaging six items, each evaluated on a six-point scale (ranging from 1, "strongly disagree", to 6, "strongly agree"). These items gauge factors such as "the clarity and number of social norms, the degree of tolerance for norm violations, and overall compliance with social norms in each nation" ([Gelfand et al., 2011](#)).

<sup>13</sup> We develop further on this below, when we present our descriptive evidence.

<sup>14</sup> For example, in a country  $c$  at calendar time *October 2015*, the dependent variable is expressed as the number of registered Golf cars (in logarithm) as of *October 2015* minus the same figure in *October 2014*. In this example,  $t = \{\text{October}, 2015\}$ ,  $\kappa = \{\text{October}\}$ , and  $m = \{\text{Golf}\}$ .

<sup>15</sup> By way of illustration, Americans' preference for larger cars, VW's clean diesel campaign running before the scandal as well as the stage of a model life cycle and discount promotions on specific models are among the most prominent examples of influential factors which are netted out by the inclusion of these terms. And so are idiosyncratic demand shocks hitting destination markets.

$$\begin{aligned}
& + \eta \cdot [Post_t \times EnvConc_c] + \theta \cdot [Post_t \times CulTight_c] + \\
& + f_{c \times \kappa} + f_{c \times m} + f_{\kappa \times m} + \epsilon_{mct}.
\end{aligned} \tag{2}$$

Our interest lies in the two triple interaction terms whose coefficients,  $\gamma^g$  and  $\xi^g$ , provide, jointly with  $\beta^g$ , key insights into the role played by *EnvConc* and *CulTight*, respectively, in modulating consumers' reactions.<sup>16</sup>

### 3.3. Treatment and control groups

To better appreciate the nuanced implications of the scandal we consider three different treatment groups throughout the paper.

The first treatment group, named “VWG Models with Defeat Device”, is defined upon a *law-infringement* criterion: it comprises the models, produced by the VWG, listed by EPA as “affected vehicles” in its Notice of Violation of the Clean Air Act to VW as per September 18th, 2015.<sup>17</sup> The EPA's listing refers exclusively to the diesel version of these models. Unfortunately, because our registry data does not specify the engine type mounted on new cars, we are compelled to define our treatment dummy at the model level. Consequently, this treatment group includes cars with gasoline engines, which were never implicated. This aggregation, if it introduces any bias, tends towards producing more conservative estimates. Thus, for this treatment group, our approach may yield an underestimation of consumers' true response to VW's deceptive conduct.<sup>18</sup>

Our second treatment group, named “All VW Models”, is defined upon the *branding* criterion: it comprises only VW-branded models, regardless of whether they have been incriminated. Our last treatment group, named “All VWG models”, is the largest one as it comprises all the models produced by all the automakers that are part of VWG. This is clearly the least refined treatment group as it tends to “tar all cars with the same brush”.

Table A.3 gives an overview of our treated groups by segments. It is worth stressing that the law-infringement criterion and the branding criterion lead to quite similar treatment groups, for segments C and D. In fact, according to EPA, all cars mounting a defeat device with the 2.0 liter diesel engine in segments C and D were, with the exception of the “Audi A3”, all VW-branded models. The heterogeneity of these two treated groups grows considerably when *higher segments* are considered. Indeed, many of the 3.0 liter diesel models produced by the VWG result in being treated according to the law-infringement criterion, but not under the branding criterion, as VW branded-models and Audi-branded models are respectively under-represented and over-represented in the EPA listing for *higher segments* (i.e., only the VW Touareg among higher class models enters the EPA list).

Moreover, to further refine our analysis, we compare our treatment groups with different control groups, each time focusing on specific segment aggregates. In the first instance, we examine how the “VWG Models with Defeat Device”, i.e., those included in treatment group 1, have fared compared to all the others (“law-complying”) models in the VWG. This provides valuable insights into the *within-group* dimension of the scandal. Similarly, “All VW Models”, i.e., those in treatment group 2, are compared to models manufactured by other VWG brands (excluding VW). Such comparisons, arguably the most “clinical” ones achievable with our data, are particularly significant as they compare “apples to apples”.

Finally, comparing the sales performance of models in the first, second and third treatment group respectively, with those of models produced by the competitors of the VWG, provides insights into the *across-groups* dimension of the scandal.

### 3.4. Descriptive evidence

We now present some descriptive evidence on how the monthly sales volume variations (our dependent variable) compare for each treatment and control group considered in our empirical analysis. In all the figures discussed in this section, the time frame spans from September 2014 to September 2016. The dashed vertical line at September 2015 marks the division between one year before and one year after the scandal.

We focus here on segments C and D as they involved the vast majority of the models implicated in the scandal and represent the object of the analysis providing our main results. Following the infringement criterion, Figs. 1(a) and 1(b) display monthly sales volume variations in segments C and D for “VWG Models with Defeat Device” (treatment group 1, red line), compared to “Other VWG Models” and “Non-VWG Models” (blue lines), respectively. While Panel (a) shows *prima facie* evidence on the *within-group* impact of the scandal, Panel (b) provides insights into the *across-groups* dimension of the scandal. In both comparisons, the negative gap between the red and blue lines widens during the initial months of 2016 and gradually narrows after about a year, revealing a drop in sales volumes of treated units. Such reaction seems somewhat delayed in relation to the outbreak of the scandal and possibly short-lived.<sup>19</sup>

Moreover, both panels exhibit parallel pre-treatment trends in the two segments most affected by the scandal. This suggests a strong indication that the public was unaware of the imminent scandal and that there were no “anticipation effects”.

<sup>16</sup> Typical for a DiD specification,  $Post \times EnvConc$  and  $Post \times CulTight$  net out potential differences in sales performance associated with average levels of environmental concern and cultural tightness before and after the scandal.

<sup>17</sup> A few other models that were found later on in some countries of our sample to have defeat devices installed are nevertheless not included in this treatment group to ease the implementation of the analysis and the interpretation of the related findings.

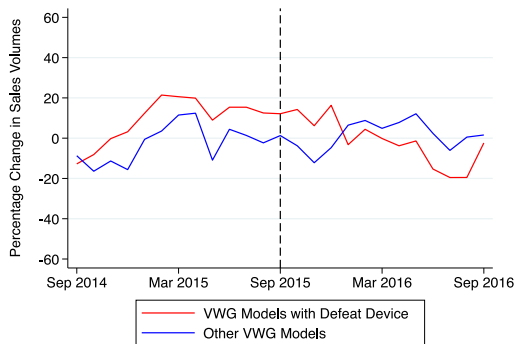
<sup>18</sup> Our argument, that we are underestimating consumer reaction, implicitly assumes that the observed decline in sales volumes is due to models equipped with the illicit device, while sales of gasoline cars were not affected by the scandal. Since we cannot observe both types of cars separately, we have to rely on a larger treatment group (which includes gasoline cars as well), and this may lead to an underestimation of the real treatment effect.

<sup>19</sup> Such delay is presumably due the unfolding of the events and to the fact a certain amount of cars had been pre-ordered before the eruption of the scandal.



(a) “Within-Group” comparison.

VWG Models with Defeat Device vs Other VWG Models



(b) “Between-Groups” comparison.

VWG Models with Defeat Device vs Non-VWG Models

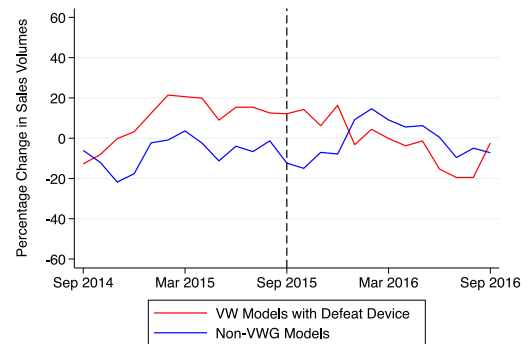
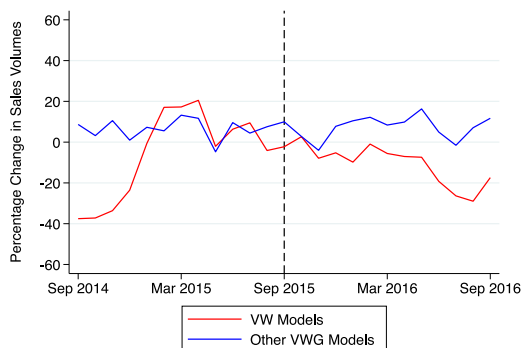


Fig. 1. Treatment Group 1 - Pre- and post-scandal monthly sales dynamics for segments C&amp;D.

(a) “Within-Group” comparison.

VW Models vs Other VWG Models



(b) “Between-Groups” comparison.

VW Models vs Non-VWG Models

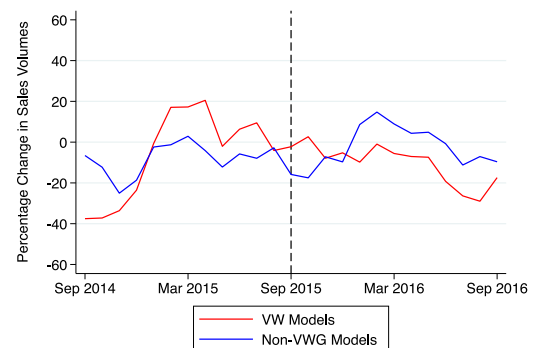


Fig. 2. Treatment Group 2 - Pre- and post-scandal monthly sales dynamics for segments C&amp;D.

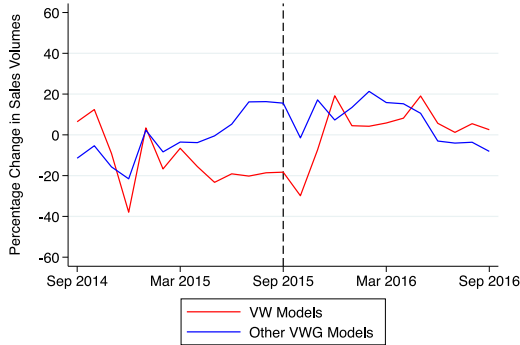
Figs. 2(a) and 2(b) concern treatment group 2 and, according to the *branding* criterion, compare VW models (red line) with other VWG models (e.g., Audi, Seat, and Škoda models) and Non-VWG models, once again for segments C and D. Examining the patterns of the red and blue lines leads to a similar conclusion: despite the aforementioned delay, the sales performance of VW's models was comparatively worse in the aftermath of the scandal.

Overall, our descriptive evidence suggests that consumers are holding VW accountable for its misconduct by refraining from purchasing the *affected* models in segments C and D. These conclusions are confirmed by our regression analysis below, where we account for unobserved confounding factors. Additionally, our regression analysis will establish the significant role of consumers' environmental concern in explaining these figures.

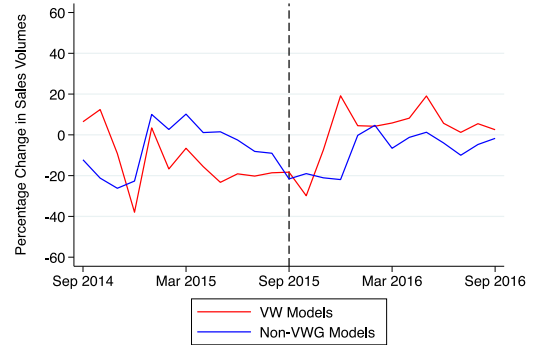
#### 4. Main results: the impact of the scandal on the implicated models

Since the effects of the scandal on VW's corporate sales are nuanced and complex, we proceed in two distinct steps. We start our analysis by examining segments C and D as they included the bulk of the models implicated in the scandal, the results are presented in this section. We first estimate Eq. (1) to evaluate the post-scandal corporate performance at the aggregate level for these segments. This provides an initial overview of how consumers globally reacted to the scandal. Subsequently, we exploit the multi-country dimension of our dataset to assess market idiosyncrasies in consumer responses to the scandal. Utilizing the augmented specification described by Eq. (2), we address the primary question of our investigation: does such heterogeneity correlate with consumers' environmental concern and cultural tightness?

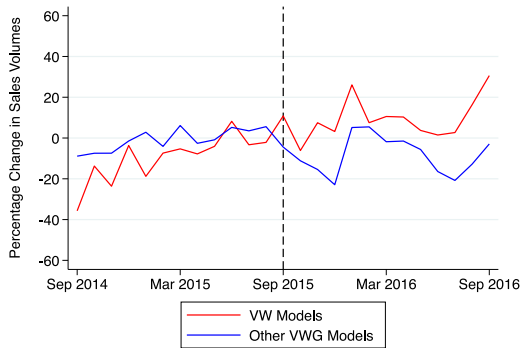
(a) *Segments A&B - “Within-Group”.*  
*VW Models vs Other VWG Models*



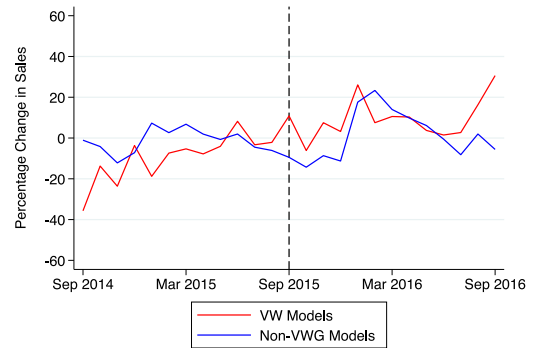
(b) *Segments A&B - “Between-Groups”.*  
*VW Models vs Non-VWG Models*



(c) *Higher Segments - “Within-Group”.*  
*VW Models vs Other VWG Models*



(d) *Higher Segments - “Between-Groups”.*  
*VW Models vs Non-VWG Models*



**Fig. 3.** Treatment group 2 - Pre- and post-scandal monthly sales dynamics by segments.

Continuing our analysis, in Section 5 we shift our focus to segments below and above segments C and D. Once more, we begin with our basic specification, aiming to understand whether adverse consumer reactions in specific segments may have been mitigated through substitution patterns within the VWG at an aggregate level. Thus, we explore whether the aforementioned substitution patterns were influenced by consumers’ environmental concern and cultural tightness.

The presentation of the estimation results consistently follows a structured pattern that aligns with the definitions of the treatment groups. Columns (i) and (ii) of the regression tables present the results referring to treatment group 1 (“VWG Models with Defeat Device”), followed by columns (iii) and (iv), reporting results obtained for treatment group 2 (“All VW models”). Finally, column (v) refers to the treatment group 3 (“All VWG models”). Each treatment group is compared with different control groups, so to assess *within-group* (columns (i) and (iii)) or *across-groups* comparisons (columns (ii), (iv) and (v)). All results presented below span over a very narrow time window (2014–2016) around the outbreak of the scandal.

#### 4.1. The direct impact

Table 4 presents the results of the estimation of Eq. (1), for segments C and D. Results are reported in two main panels. The upper one encompasses all car producers available in our data set (cf. Table A.2 in the appendix). The lower one (“Non-Alleged Groups”)



**Table 4**  
Scandal effect — Segments C & D.

| Segments C & D (Sep 2014–Sep 2016)                                |                                     |                                     |                        |                      |                     |
|-------------------------------------------------------------------|-------------------------------------|-------------------------------------|------------------------|----------------------|---------------------|
|                                                                   | Treatment group 1                   |                                     | Treatment group 2      |                      | Treatment group 3   |
|                                                                   | <within-group>                      | <across-group>                      | <within-group>         | <across-groups>      | <across-groups>     |
|                                                                   | (i)                                 | (ii)                                | (iii)                  | (iv)                 | (v)                 |
| <i>Treated Units</i>                                              | VWG<br>Models with<br>Defeat Device | VWG<br>Models with<br>Defeat Device | All VW<br>Models       | All VW<br>Models     | All VWG<br>Models   |
| <i>TreatXPost</i>                                                 | −0.1265***<br>(0.0445)              | −0.0273<br>(0.0758)                 | −0.1297***<br>(0.0442) | −0.0031<br>(0.0830)  | 0.0774<br>(0.0828)  |
| <i>Control Units</i>                                              | Other VWG<br>Models                 | Non-VWG<br>Models                   | Other VWG<br>Models    | Non-VWG<br>Models    | Non-VWG<br>Models   |
| <i>Observations</i>                                               | 6,380                               | 30,741                              | 6,380                  | 27,486               | 31,359              |
| <i>R-squared</i>                                                  | 0.2906                              | 0.4129                              | 0.2906                 | 0.4301               | 0.4127              |
| Segments C & D - Excluding “Alleged” Groups (Sep 2014 - Sep 2016) |                                     |                                     |                        |                      |                     |
|                                                                   | Treatment group 1                   |                                     | Treatment group 2      |                      | Treatment group 3   |
|                                                                   | <within-group>                      | <across-group>                      | <within-group>         | <across-groups>      | <across-groups>     |
|                                                                   | (i)                                 | (ii)                                | (iii)                  | (iv)                 | (v)                 |
| <i>Treated Units</i>                                              | VWG<br>Models with<br>Defeat Device | VWG<br>Models with<br>Defeat Device | All VW<br>Models       | All VW<br>Models     | All VWG<br>Models   |
| <i>TreatXPost</i>                                                 | −0.1265***<br>(0.0445)              | −0.1242**<br>(0.0582)               | −0.1297***<br>(0.0442) | −0.1196*<br>(0.0586) | −0.0316<br>(0.0676) |
| <i>Control Units</i>                                              | Other VWG<br>Models                 | Non-VWG<br>Models                   | Other VWG<br>Models    | Non-VWG<br>Models    | Non-VWG<br>Models   |
| <i>Observations</i>                                               | 6,380                               | 23,549                              | 6,380                  | 20,294               | 24,167              |
| <i>R-squared</i>                                                  | 0.2906                              | 0.3967                              | 0.2906                 | 0.4179               | 0.3959              |

*Note:* The dependent variable is the change in the monthly sales volumes of a car-model registered in a country. This change is measured as the first difference in the logarithm of sales volumes (on the corresponding month of the previous year). Standard errors are robust and clustered at the group level in columns (ii), (iv) and (v). All specifications include a post-treatment indicator, along with country-month, country-model and month-model fixed effects. Segments C and D (*medium* and *large* cars) are segment classes of the European passenger car classification. The set of “Alleged” Groups includes BMW Group, Daimler Group, FCA, Renault-Nissan Alliance. \*, \*\*, and \*\*\* denote significance at the 10, 5, and 1 percentage level, respectively.

excludes from the sample automakers that, subsequently to the VW scandal, have also been under investigation for environmental-law infringement.<sup>20</sup> Thus, this panel probably offers the cleanest possible comparison for the *across-groups* comparison, as the control group comprises exclusively cars manufactured by groups free of allegations of environmental misconduct.

The *within-group* comparisons are introduced in columns (i) and (iii) of Table 4 and reveal a punitive response of consumers to VW's corporate behavior in our time window. Indeed, sales of cars mounting a defeat device and belonging to the VWG as well as purchases of VW-branded models decreased by about 12.6 percentage points, when compared to other models of the VWG in the same segment.<sup>21</sup> Across all panels, our estimates are robust both in terms of the estimated magnitudes and significance level: the very similar coefficient found for columns (i) and (iii) is also reassuring because, as noted above, the two treatment groups represented in these two columns are quite similar and comparable for segments C and D.

When it comes to the *across-groups* dimension of the scandal, findings are less clear cut. Estimates reported in columns (ii) and (iv) are significant only in the lower panel where the “alleged” groups are excluded from the sample. They indicate that sales volumes of VWG models with the defeat device (as much of VW models) have decreased by roughly 12 percentage points, if compared to vehicles produced by competitors in the same segments. Column (v) refers to the third treatment group, “All VWG models”. Here, we do not find evidence of a significant negative reaction, suggesting that VWG as a whole has not been punished by consumers, even in the segments most directly involved in the scandal.

The analysis of segments C and D yields a few insights. First, customer backlash against the VWG, has mainly targeted models directly implicated in the scandal, the large share of which are VW-branded models. However, this negative consumer response does

<sup>20</sup> In the months following the VW emissions scandal other car manufacturers have been accused of altering the test results of their cars. For the results showed in the lower panel of Table 4, we removed Daimler Group, BMW Group, FCA and Renault-Nissan Alliance from the control group as they have also been investigated for cheating at the emissions tests.

<sup>21</sup> To avoid any source of possible confusion, it is worth noting that the composition of treated and control groups slightly differ in columns (i) and (iii). Although both columns reflect a *within-group* comparison and include only models in segments C and D, the control group in column (i) consists of all other models in the VWG without a *defeat device*, whereas in column (iii) the control group comprises all non-VW cars produced by the VWG. Therefore the expression “other VWG models” is to be understood always in relation to the complementary set implied by the treated group within the boundary of the VWG and within the boundary of the segment considered.

Table 5

Drivers – environmental concern vs Cultural tightness – segments C &amp; D.

|                            | Segments C & D                              |                                     |                        |                        |                       |
|----------------------------|---------------------------------------------|-------------------------------------|------------------------|------------------------|-----------------------|
|                            | Treatment group 1                           |                                     | Treatment group 2      |                        | Treatment group 3     |
|                            | <within-group>                              | <across-groups>                     | <within-group>         | <across-groups>        | <across-groups>       |
|                            | (i)                                         | (ii)                                | (iii)                  | (iv)                   | (v)                   |
| <i>Treated Units</i>       | VWG<br>Models with<br>Defeat Device         | VWG<br>Models with<br>Defeat Device | All VW<br>Models       | All VW<br>Models       | All VWG<br>Models     |
| <i>TreatXPost</i>          | −0.0022<br>(0.0940)                         | 0.2211*<br>(0.1181)                 | 0.1124<br>(0.0950)     | 0.3272**<br>(0.1327)   | 0.2525*<br>(0.1325)   |
| <i>TreatXPostXEnvConc</i>  | −0.4959***<br>(0.1491)                      | −0.5332***<br>(0.0891)              | −0.6175***<br>(0.1470) | −0.5581***<br>(0.0990) | −0.2155**<br>(0.0989) |
| <i>TreatXPostXCulTight</i> | 0.1452<br>(0.1518)                          | −0.1150<br>(0.0954)                 | −0.0052<br>(0.1532)    | −0.2661**<br>(0.1016)  | −0.2305**<br>(0.1023) |
| <i>Control Units</i>       | Other VWG<br>Models                         | Non-VWG<br>Models                   | Other VWG<br>Models    | Non-VWG<br>Models      | Non-VWG<br>Models     |
| <i>Observations</i>        | 6,380                                       | 30,741                              | 6,380                  | 27,486                 | 31,359                |
| <i>R-squared</i>           | 0.2953                                      | 0.4152                              | 0.2960                 | 0.4309                 | 0.4133                |
|                            | Segments C & D - Excluding “Alleged” Groups |                                     |                        |                        |                       |
|                            | Treatment group 1                           |                                     | Treatment group 2      |                        | Treatment group 3     |
|                            | <within-group>                              | <across-groups>                     | <within-group>         | <across-groups>        | <across-groups>       |
|                            | (i)                                         | (ii)                                | (iii)                  | (iv)                   | (v)                   |
| <i>Treated Units</i>       | VWG<br>Models with<br>Defeat Device         | VWG<br>Models with<br>Defeat Device | All VW<br>Models       | All VW<br>Models       | All VWG<br>Models     |
| <i>TreatXPost</i>          | −0.0022<br>(0.0940)                         | 0.0843<br>(0.1048)                  | 0.1124<br>(0.0950)     | 0.1729<br>(0.1278)     | 0.0984<br>(0.1276)    |
| <i>TreatXPostXEnvConc</i>  | −0.4959***<br>(0.1491)                      | −0.4526***<br>(0.1154)              | −0.6175***<br>(0.1470) | −0.4453***<br>(0.1217) | −0.1029<br>(0.1218)   |
| <i>TreatXPostXCulTight</i> | 0.1452<br>(0.1518)                          | −0.0944<br>(0.0865)                 | −0.0052<br>(0.1532)    | −0.2585**<br>(0.0907)  | −0.2234**<br>(0.0911) |
| <i>Control Units</i>       | Other VWG<br>Models                         | Non-VWG<br>Models                   | Other VWG<br>Models    | Non-VWG<br>Models      | Non-VWG<br>Models     |
| <i>Observations</i>        | 6,380                                       | 23,549                              | 6,380                  | 20,294                 | 24,167                |
| <i>R-squared</i>           | 0.2953                                      | 0.3978                              | 0.2960                 | 0.4195                 | 0.3971                |

Note: The dependent variable is the change in the monthly sales volumes of a car-model registered in a country. This change is measured as the first difference in the logarithm of sales volumes (on the corresponding month of the previous year). The variable “*EnvConc*” is the *environmental concern index* expressing customers’ “environmental concern” in a unit-scale. The variable “*CulTight*” is the *cultural tightness index* measuring the *pervasiveness of social norms* in a country on a unit-scale. Standard errors are robust and clustered at the group level in columns (ii), (iv) and (v). All specifications include a post-treatment indicator and its interactions with *EnvConc* and *CulTight*, alongside country-month, country-model, and month-model fixed effects. Segments C and D (*medium and large cars*) are segment classes of the European passenger car classification. The set of “Alleged” Groups includes BMW Group, Daimler Group, FCA, Renault-Nissan Alliance. \*, \*\*, and \*\*\* denote significance at the 10, 5, and 1 percentage level, respectively.

not translate into a loss in sales at the VWG level. This suggests that consumers have, to a certain extent, redirected their preferences to other models, in the same segments, within the VWG.

Second, VW’s forgone sales in relation to other automaker groups - the *across-groups* effect described above - become apparent when focusing on a specific subset of competitors – those groups untainted by allegations of environmental misconduct.

Finally, we have also extended the time horizon of our analysis to the whole sample period, 2013–2017.<sup>22</sup> As we extend our time window, the results from the *within-group* comparison weaken significantly. From this we deduce that the punishment effect that we have identified is relatively *short-lived*, with its effect not extending beyond the year 2016.

#### 4.2. Environmental concern as a driver of consumer behavior

A compelling feature of our data is that we observe car registrations in multiple countries. This enables us to explore the extent to which the reaction of consumers, concerning models in segments C and D, differs across national markets. In order to have an overview of such heterogeneity, and before looking at its relation with environmental concern or cultural tightness, we rely on a modified version of regression (1) for models in segments C and D, in which the treatment variable – the pool of VWG’s models whose diesel version mounted a defeat device – is interacted with each individual country dummy. The results are reported in Fig. 4.

<sup>22</sup> Results can be made available upon request.

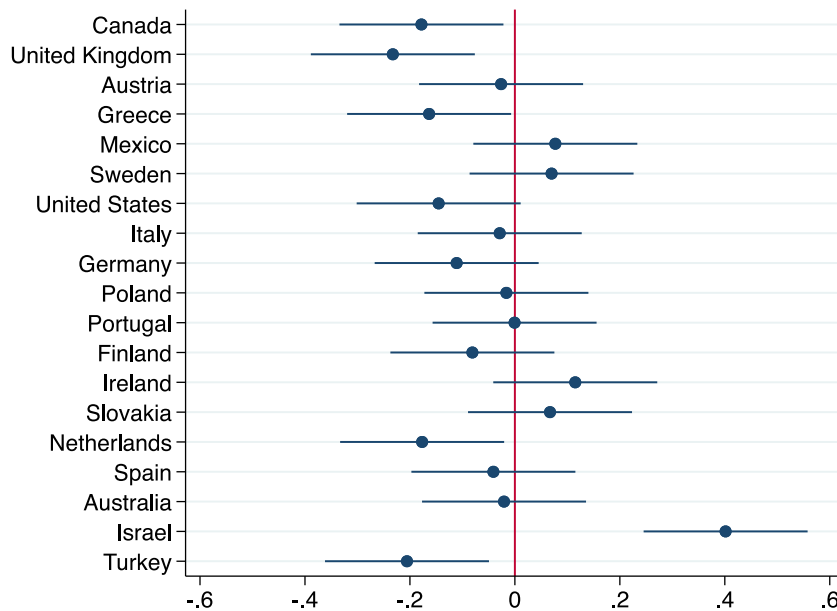


Fig. 4. Market-idiosyncrasies: Penalty of VWG's models with a defeat device

Note: The horizontal axis displays the point estimate along with the confidence interval associated to individual sales markets in a modified version of specification (1), in which the treatment effect is interacted with single country dummies. The vertical axis (bottom to top) lists countries in increasing order of their environmental concern index.

The graph plots, on the horizontal axis, the point estimates along with the confidence interval bands for each country obtained from this regression.<sup>23</sup>

A substantial market heterogeneity is readily apparent, with Anglo-Saxon markets like UK, Canada, and partly US, registering among the most pronounced consumer penalties. In a handful of European markets, namely Greece, Germany, Netherlands, and Turkey, the decline in sales volumes of EPA-listed models also exceeds the pooled average. In other parts of the globe, consumers' reaction looks rather tame, with forgone sales on these models being minimal and statistically negligible. However, this figure unveils one dimension that an average effect conceals: affected models were even performing comparatively better than models produced by rival groups in specific markets — most notably in Israel and, to a lesser extent, Ireland.

Having documented market idiosyncrasies in consumers' initial responses to the scandal, we delve next into potential underlying explanations for their existence. We commence with the premise that culture, social norms, and institutional factors may influence economic decisions. We are able to associate consumer reactions to the scandal with their environmental concern and their attitude towards social norms. As mentioned above, we expect that a stronger environmental concern among consumers in a country, along with a higher level of cultural tightness, should trigger more punitive reactions to the scandal.

Fig. 4, introduced earlier, lends support to our prior. Countries are purposely ranked according to their ascending level of environmental concern, along the vertical axis namely, environmental concern increases as one moves from the bottom to the top. Eyeball evidence shows that sales volumes of cars featuring defeat devices tend to decline as the level of environmental concern increases.<sup>24</sup>

We put our conjectures to the test by estimating Eq. (2), which includes the variable  $Treat \times Post$  interacted with our *environmental concern* and *cultural tightness* measures, respectively. If our hypotheses hold true, we expect that a more pronounced punitive effect associated with the scandal, that is, a negative (and statistically significant) coefficient associated with either of the two triple interaction terms.<sup>25</sup>

As shown in Table 5, the triple interaction terms just introduced exhibit predominantly a negative sign, indicating that both channels operate in the expected direction. This is true both for VWG models with a defeat device as well as VW branded models.

<sup>23</sup> For clarity, the control group in this regression consists of the complementary set to the treated group — that is, all other models mounting no defeat device. As such, it has no exact counterpart to the control groups used in our regression tables. Given the purely descriptive and illustrative nature of this regression, this choice is motivated merely by statistical efficiency: exploiting the whole sample of observations enables to retrieve sensible confidence intervals for individual countries, notwithstanding our demanding specification.

<sup>24</sup> There are of course a few notable “non-compliers” with our hypothesis — Turkey, Mexico, Sweden, and the Netherlands being the most prominent — for which the linear fit of our regression will likely be weaker.

<sup>25</sup> The scale of our indexes eases the interpretation of the coefficients associated to our triple interaction term. Specifically, the lowest ranked market (on each scale) serves as the reference market. Similarly, the highest-ranked country yields a triple interaction term of one. The interpretation of our findings remains also consistent when using standardized indexes instead of normalized ones.

However, only the coefficient associated to the environmental concern channel results statistically significant across specifications. It follows that most of the punishment effect shown in Table 4 can likely be attributed to responses from environmentally concerned consumers. Their influence is far from negligible: holding other factors constant, one may reasonably expect an additional decrease in sales volumes of roughly 50 percentage points, when comparing the country with the highest level of environmental concern with the country with the lowest one.<sup>26</sup> The quantitative implications of the coefficients appearing in Table 5 are best illustrated through the example of Greece and Spain using the coefficient in column (ii). Despite sharing the same cultural tightness index of 0.2, they diverge in environmental concern levels: 0.81 for the former and 0.14 for the latter. This discrepancy translates into an approximate 35 percentage points additional sales decline in the *across-groups* comparison.<sup>27</sup> A similar decline is implied also by the *within-group* comparison shown in column (i).

As for cultural tightness, which serves in our context as a metric for societal adherence to social norms and values, our results find hardly support in favor of its influential role in channeling consumer reactions to corporate misconduct into reduced purchases of VWG cars implicated in the scandal.

Therefore, in the comparative assessment between environmental concern and cultural tightness, it is evident that environmental concern emerges as a key factor in steering consumer post-scandal reactions in many of the markets included in our sample.

#### 4.3. Sensitivity analysis

To corroborate our main findings we run a few robustness checks.

First, the conclusions about the role of environmental concern remain substantially unchanged also when utilizing the *environmental concern* measure constructed by Franzen and Vogl (2013), available for a narrower set of countries (see Table A.4).

Second, the unique and idiosyncratic consumer reaction observed in the Israeli market following the Volkswagen emissions scandal, as shown in Fig. 4, prompted us to conduct a specific robustness check. This was to ensure that our results were not solely driven by this particular finding. As reported in Table A.5, excluding observations from the Israeli market did not substantially alter our main estimates. These remain closely aligned with those illustrated in Table 4. Additionally, as presented in Table A.6, the distinct characteristics of the Israeli market did not significantly impact the relevance of the environmental concern channel. Once again, our estimates closely resemble those obtained from the full sample.

Finally, we have taken additional steps to enhance the robustness of our analysis by addressing the potential influence of countries (Italy, the Netherlands, Spain, the United Kingdom, and the United States) where the sales of the implicated cars were temporarily halted after the scandal came to light. Through the exclusion of these countries from our analysis, our aim is to rule out the impact that temporary bans might have had, and to evaluate whether they affected or interacted with environmental concern and cultural tightness in shaping consumer reactions. The findings from this refined analysis, as presented in Table A.7, indicate that bans or analogous interventions did not significantly influence consumer reactions and their link with the level of environmental concern.

#### 5. Consumer reaction and substitution patterns

Our analysis has centered so far only on segments C and D, that included most of the models implicated in the scandal. We turn now our attention also to other segments for a more nuanced analysis of the reaction of consumers to the scandal. The benefit of a broader analysis is twofold: first, it permits to gauge the *pervasiveness* of the effects of the scandal inside the VWG. Moreover, comparing different treatment effects obtained from different segment aggregates, we can gain insights into potential substitution patterns triggered by the scandal, despite the absence of price and discount data at the transaction level.

This idea builds on the following premise. Cars in segments A and B, particularly those in segment B, can be regarded as substitutes for consumers who are on the brink of purchasing a vehicle from segment C. As an example, a potential consumer willing to buy a VW car in segment-C (e.g., a VW Golf) may be prompted by the scandal to change her preference in favor of a different VW model in a lower segment (e.g., a VW Polo in segment B), leading to increased VW sales in this lower segment.<sup>28</sup> If this behavior is prevalent in the data, the finding that VW's sales in segments A and B outperform the sales of other brands within the VWG (or of brands belonging to competing groups) in the same segments (i.e., a positive treatment effect), alongside a negative impact on segments C and D, would be consistent with a substitution dynamic within the VW brand between different segments. Likewise, a consumer willing to buy a segment-D VW car (e.g., Passat) may be prompted by the scandal to choose a VW model from a higher segment (e.g., a VW Touran). This behavior is consistent with an increase in the sales in higher segments at the expense of sales in segment D.

Generalizing from these examples, detecting *within-brand* substitutions (e.g., Polo for Golf or Touran for Passat) requires in our setting comparing the *within-group* effects between different segments. Likewise, detecting leakages of consumers away from VW brand or VWG group necessitates contrasting the *across-groups* effects between different segments. Building upon this idea in the next sub-section we re-estimate Eqs. (1) and (2) for segments A and B and *higher segments*, respectively.<sup>29</sup>

<sup>26</sup> For brevity, we abstain from presenting regressions based on *all segments*, which can be made available upon request. In this case, the estimated effect is slightly smaller.

<sup>27</sup> This is calculated as follows:  $-0.53 \times (0.81 - 0.14) = -0.35$ , where  $-0.53$  is the coefficient associated to the regressor " $Treat \times Post \times EnvConc$ " in the upper panel of Table 5, column(ii).

<sup>28</sup> If the scandal induces the consumer to change her preference in favor of a car from another brand within the VWG, in the same segment (e.g., a Škoda Octavia), this type of substitution is already seized by the treatment effect obtained above from the analysis of segments C and D.

<sup>29</sup> Given the limitation of our data, caution will be needed to interpret this type of evidence. We speak – in fact – of “hints” of substitution patterns because our current setting allows us at most to validate or reject the null hypothesis of “punishment” at the segment level. Substitution *within VW brand* (between

### 5.1. Substitution within or away from the VW brand

Before we look into our empirical results it may be useful to present some descriptive evidence, summarized in Figs. 3(a) to 3(d), that offer some valuable insights into the occurrence of both within-group and across-groups substitution effects between segment categories. This involves jointly examining the plots for segments C and D (depicted already in Figs. 2(a) and 2(b)) alongside those for segments A and B, as shown in Figs. 3(a) and 3(b), or higher segments, as reported in Figs. 3(c) and 3(d). Analyzing the trends within segments A and B, we observe a notable surge in sales volumes for VW models, both when compared to other VWG models and non-VWG models.

This evidence, jointly with what is shown for segments C and D, suggests a shift in consumer preference towards lower-segment VW-branded cars, rather than towards models of other brands within the group or different automotive groups altogether. These trends are less pronounced for higher-segment models.

We can now turn to the empirical evidence starting with the results on the pooled sample. Because EPA's official announcement does not contain any vehicle belonging to segments A and B and VW-branded models belonging to higher segments are less represented among the EPA-listed models, our definition of treatment group 1, based on the presence of the defeat device ascertained by EPA, becomes inapplicable or impracticable outside segments C and D. Therefore, treatment group 1 does not enter Table 6: estimates for segments A and B in the upper panel, and for higher segments in the lower panel, refer only to treatment group 2 - columns (iii) and (iv) - and treatment group 3 - column (v). Additionally – for conciseness – we present results exclusively from the sample of the “non-alleged” groups.

Let us first focus on column (iii), which refers to the *within-group* dimension of the scandal. In both panels, the estimated coefficient is positive, suggesting no discernible penalty for VW models belonging to these segments. However, the coefficient is only significant at 1% level for higher segment models. In conjunction with the results obtained above, for segments C and D, we interpret these findings as follows. Our data illustrate the possibility that foregone sales volumes experienced in segments C and D might have been, to some extent, counterbalanced by *within-brand* substitution: certain consumers may have been prompted by the scandal to choose a VW model belonging to higher segments (e.g., a VW Touran) instead of a VW model in segments C and D (e.g., a VW Passat). Albeit insignificant, the positive sign of the coefficient in column (iii) suggests nevertheless a similar *within-brand* type of substitution between cars in segments C and D (e.g., VW Golf) with those in the lower segments A and B (e.g., VW Polo).

Moreover, the positive and significant coefficient illustrated in column (iv), based on a control group comprising models produced by VWG's competitors (from “non-alleged” manufactures), provides further clues of *within-brand* substitution towards both higher and lower segments, showing also that there has not been a leakage of customers to rival models.<sup>30</sup>

Above all, these figures well illustrate that sales volumes of VW models in segments A and B, as well as in higher segments, have played a role in mitigating the decline in sales observed for models in segments C and D. This has presumably enabled VW, along with its consolidated holdings, to navigate the sales downturn stemming from segments C and D more adeptly in the wake of the scandal.

Our last goal is to determine whether the substitution patterns observed at the aggregate level are also related to the varying consumer responses across countries. Additionally, we aim to explore if this heterogeneity can be linked to the countries' levels of environmental concern or to their degree of cultural tightness. To do so we rely once again on the augmented specification (2), for segments A and B, and for higher segments, respectively. The results are shown in Table 7. The lower panel dedicated to higher segments indicates that our findings do not confute the possibility of environmentally concerned consumers displaying a greater inclination to substitute VW and VWG higher-end models with those produced by competitors. This inference arises from the negative and statistically significant coefficient on the triple interaction term “ $Treat \times Post \times EnvConc$ ”.

Conversely, environmentally concerned consumers appear to have potentially substituted VW models in segments C and D with other VW models in the lower segments A and B, allowing VW to fare better in this segment than its non-alleged competitors and its own subsidiary brands of the VWG (cf. column (iii) and (iv), upper panel of Table 7). This suggests that the lower-end segments have likely played an important role in dimming the effects of the crisis, particularly in markets more sensitive to environmental misconduct. Indeed, while the decline in sales volumes for segment C and D sees an additional reduction of 50-60 percentage points in the most environmentally conscious markets, sales performance in the lowest segments A and B seem to experience in the same markets a comparable increase.

Finally, the nuances and complexity just described go largely unnoticed were one to rely exclusively on the comparison offered in column (v), in which VWG's corporate sales performance is contrasted with those of other automotive groups. This once again underlines the limitations and potential pitfalls in the approach of “tarring all cars with the same brush”. Furthermore, when the effect in column (v) aligns with the ones reported in the other columns, its magnitude is more than halved.

In conclusion, in the light of an extremely demanding specification in accounting for unobserved heterogeneity, we find a strong support for – what we label – the *environmental concern channel*. This suggests that consumers' punishment regarding Volkswagen's deceptive practices is largely rooted in their environmental attitude.

segment categories) and consumers' leakage away from VW brand or VWG are viable possibilities that cannot be rejected with the data at hand, but they are also not the only factor concurring to shape the data in the same way.

<sup>30</sup> This is, however, not to deny consumers' leakage away from VW cars in segments C and D and towards VWG's competitors' cars in higher segments as viable. This type of substitution may in fact be concomitantly happening, but it remains undetected in our data as it does not translate into a sales growth differential in favor of other manufactures immediately after the scandal.

**Table 6**  
Scandal effect — Segments A & B and higher segments.

|                         | Segments A & B - Excluding “Alleged” Groups  |                       |                       |
|-------------------------|----------------------------------------------|-----------------------|-----------------------|
|                         | Treatment group 2                            |                       | Treatment group 3     |
|                         | <within-group>                               | <across-groups>       | <across-groups>       |
|                         | (iii)                                        | (iv)                  | (v)                   |
| <i>Treated Units</i>    | All VW Models                                | All VW Models         | All VWG Models        |
| <i>Treatment effect</i> | 0.0276<br>(0.0609)                           | 0.4469***<br>(0.1213) | 0.4286***<br>(0.1212) |
| <i>Control Units</i>    | Other VWG Models                             | Non-VWG Models        | Non-VWG Models        |
| <i>Observations</i>     | 2,645                                        | 10,318                | 12,143                |
| <i>R-squared</i>        | 0.2503                                       | 0.5218                | 0.5049                |
|                         | Higher Segments — Excluding “Alleged” Groups |                       |                       |
|                         | Treatment group 2                            |                       | Treatment group 3     |
|                         | <within-group>                               | <across-groups>       | <across-groups>       |
|                         | (iii)                                        | (iv)                  | (v)                   |
| <i>Treated Units</i>    | All VW Models                                | All VW Models         | All VWG Models        |
| <i>Treatment effect</i> | 0.2752***<br>(0.0455)                        | 0.2591***<br>(0.0592) | 0.0283<br>(0.0593)    |
| <i>Control Units</i>    | Other VWG Models                             | Non-VWG Models        | Non-VWG Models        |
| <i>Observations</i>     | 8,431                                        | 27,771                | 34,603                |
| <i>R-squared</i>        | 0.3049                                       | 0.4626                | 0.4384                |

*Note:* The dependent variable is the change in the monthly sales volumes of a car-model registered in a country. This change is measured as the first difference in the logarithm of sales volumes (on the corresponding month of the previous year). Standard errors are robust and clustered at the group level in columns (iv) and (v). All specifications include a post-treatment indicator, along with country-month, country-model and month-model fixed effects. Car segments follow the European passenger car classification: Segments A and B (*mini* and *small* cars); *Higher Segments* is an aggregate of segment classes (segments E, F, MPV, and SUV). The set of “Alleged” Groups includes BMW Group, Daimler Group, FCA, Renault-Nissan Alliance. \*, \*\*, and \*\*\* denote significance at the 10, 5, and 1 percentage level, respectively.

## 6. Conclusions

This study brings new evidence on the far-reaching implications of the Volkswagen emissions scandal exploiting novel data on new car registrations over 19 countries. Our main contribution lies in the comprehensive evaluation of the idiosyncratic responses triggered by the scandal across diverse countries and the extent to which their differences are motivated by consumers' environmental and cultural traits.

A first crucial insight gleaned from our work underscores the necessity of a deep analysis of sales performance in comprehending the far-reaching ramifications of Volkswagen's crisis. It becomes evident that evaluating corporate sales solely at the group level masks essential nuances like the *within-group* dimension of the scandal.

Another important achievement of our study is to provide evidence of the considerable variability in consumer reactions across markets. Our results highlight the pivotal role of environmental concern in driving this heterogeneity whereas the same is not true for cultural tightness. This interpretation gains further weight from the findings that Volkswagen performs notably better than its competitors in segments A and B, as well as in segments above C and D. This observation seems difficult to reconcile with a significant erosion of Volkswagen's customer base. Such erosion would presumably have led to a widespread sales decline across all segments. Our findings rather align with the hypothesis that consumers, and environmentally concerned consumers alike, deliberately avoided models non-compliant with the emission standards.

Finally, our results highlight that the effects of the emissions scandal were relatively confined and transient. Although consumers' punitive response to Volkswagen's corporate deceitful behavior was rooted in their environmental concern, it was nonetheless short-lived, lasting no longer than a year from the outbreak of the scandal and narrowly targeted to segments C and D. Moreover, while the scandal imposed significant financial burdens, the operating losses resulting from forgone sales could be partially offset through substitution dynamics within the brand or across the group. This is especially true in more environmentally sensitive markets, where environmentally conscious consumers have redirected their purchases within the VW brand or group, avoiding leaks to competitors' models.

Ultimately, our findings offer an additional perspective on Volkswagen's new course centered on the pursuit of the *electric vehicle transition*. As this transformation is particularly resonating with environmentally concerned consumers, it offers a plausible



**Table 7**  
Drivers – Environmental concern vs cultural tightness – Segments A & B and higher segments.

|                            | Segments A & B - Excluding “Alleged” Groups  |                        |                      |
|----------------------------|----------------------------------------------|------------------------|----------------------|
|                            | Treatment group 2                            |                        | Treatment group 3    |
|                            | <within-group>                               | <across-groups>        | <across-groups>      |
|                            | (iii)                                        | (iv)                   | (v)                  |
| <i>Treated Units</i>       | All VW Models                                | All VW Models          | All VWG Models       |
| <i>TreatXPost</i>          | −0.2543**<br>(0.1273)                        | 0.1141<br>(0.2174)     | 0.2997<br>(0.2180)   |
| <i>TreatXPostXEnvConc</i>  | 0.6502***<br>(0.2193)                        | 0.5718**<br>(0.2641)   | 0.1199<br>(0.2682)   |
| <i>TreatXPostXCulTight</i> | 0.1030<br>(0.2071)                           | 0.2729<br>(0.3305)     | 0.1924<br>(0.3287)   |
| <i>Control Units</i>       | Other VWG Models                             | Non-VWG Models         | Non-VWG Models       |
| <i>Observations</i>        | 2,645                                        | 10,318                 | 12,143               |
| <i>R-squared</i>           | 0.2539                                       | 0.5229                 | 0.5058               |
|                            | Higher Segments - Excluding “Alleged” Groups |                        |                      |
|                            | Treatment group 2                            |                        | Treatment group 3    |
|                            | <within-group>                               | <across-groups>        | <across-groups>      |
|                            | (iii)                                        | (iv)                   | (v)                  |
| <i>Treated Units</i>       | All VW Models                                | All VW Models          | All VWG Models       |
| <i>TreatXPost</i>          | 0.4801***<br>(0.0958)                        | 0.5394***<br>(0.0982)  | 0.1461<br>(0.0980)   |
| <i>TreatXPostXEnvConc</i>  | −0.4027**<br>(0.0958)                        | −0.4875***<br>(0.0765) | −0.1391*<br>(0.0764) |
| <i>TreatXPostXCulTight</i> | −0.1051<br>(0.1546)                          | −0.2329**<br>(0.0793)  | −0.1575*<br>(0.0783) |
| <i>Control Units</i>       | Other VWG Models                             | Non-VWG Models         | Non-VWG Models       |
| <i>Observations</i>        | 8,431                                        | 27,771                 | 34,603               |
| <i>R-squared</i>           | 0.3065                                       | 0.4629                 | 0.4386               |

*Note:* The dependent variable is the change in the monthly sales volumes of a car-model registered in a country. This change is measured as the first difference in the logarithm of sales volumes (on the corresponding month of the previous year). The variable “*EnvConc*” is the *environmental concern index* expressing customers’ “environmental concern” in a unit-scale. The variable “*CulTight*” is the *cultural tightness index* measuring the *pervasiveness of social norms* in a country on a unit-scale. Standard errors are robust and clustered at the group level in columns (iv) and (v). All specifications include a post-treatment indicator and its interactions with *EnvConc* and *CulTight*, alongside country-month, country-model, and month-model fixed effects. Car segments follow the European passenger car classification: Segments A and B (*mini* and *small* cars); *Higher Segments* is an aggregate of segment classes (segments E, F, MPV, and SUV). The set of “Alleged” Groups includes BMW Group, Daimler Group, FCA, Renault-Nissan Alliance. \*, \*\*, and \*\*\* denote significance at the 10, 5, and 1 percentage level, respectively.

explanation as for the trust between Volkswagen and its customer base not being irreparably shattered in the wake of the scandal. Certainly, pursuing the merits of other explanations, for instance related to Volkswagen’s communication strategy and its endeavor in reestablishing its corporate image, remains a compelling avenue for future research, albeit beyond the scope of this paper.

### CRedit authorship contribution statement

**M. Marino:** Writing – review & editing, Methodology, Formal analysis, Conceptualization. **P. Parrotta:** Writing – review & editing, Methodology, Investigation, Formal analysis, Conceptualization. **D. Sala:** Writing – review & editing, Writing – original draft, Conceptualization. **G. Valletta:** Writing – review & editing, Writing – original draft, Investigation, Conceptualization.

### Appendix

See [Tables A.1–A.7](#)

**Table A.1**

Car segments — European passenger car classification (EC, 1999).

| Segment                             | Examples                                                                                                     |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <i>A-Mini cars</i>                  | Chevrolet Spark, Fiat 500, Kia Picanto, Suzuki Alto, Renault Twingo                                          |
| <i>B-Small cars</i>                 | Ford Fiesta, Kia Rio, Opel Corsa, Peugeot 208, Volkswagen Polo                                               |
| <i>C-Medium cars</i>                | Audi A3, BMW 1 Series, Lexus CT, Mercedes-Benz A-Class, Ford Focus, Toyota Corolla, Volkswagen Golf          |
| <i>D-Large cars</i>                 | Audi A4, BMW 3 Series, Lexus IS, Mazda 6, Mercedes-Benz C-Class, Ford Mondeo, Peugeot 508, Volkswagen Passat |
| <i>E-Executive cars</i>             | Audi A6, BMW 5 Series, Mercedes-Benz E-Class, Volvo S90                                                      |
| <i>F-Luxury cars</i>                | Audi A8, BMW 7 Series, Jaguar XJ, Mercedes-Benz S-Class, Porsche Panamera                                    |
| <i>MPV - Multi-purpose vehicles</i> | Ford C-Max, Kia Carnival, Renault Espace, Volkswagen Touran                                                  |
| <i>SUV - Sport utility vehicles</i> | Jeep Grand Cherokee, Ford Escape, Honda CR-V, Kia Sportage, Toyota RAV4, Volkswagen Touareg, Volvo XC90      |

*Note:* The car classification used in North America is somehow different from the classification used in Europe. Segments A and B are sometimes referred as *microcars* (e.g., Fiat 500) and *subcompact* (e.g., Ford Fiesta) car categories. Segments C and D find their equivalent in the *compact* (e.g., Golf), *mid-size* (e.g., VW Passat), *entry-level luxury* (e.g., Audi A4) car categories of the US classification. The superior segments are analogous to American *executive*, *mid-* and *full-size luxury* cars, *sport* cars, and *SUV*.

**Table A.2**

Sample composition by group (September 2014–September 2016).

| Group                   | Segments C & D |        | Segments A & B |        | Higher segments |        | All segments |        |
|-------------------------|----------------|--------|----------------|--------|-----------------|--------|--------------|--------|
|                         | Obs            | %      | Obs            | %      | Obs             | %      | Obs          | %      |
| Aston Martin            | –              | –      | –              | –      | 193             | 0.36   | 193          | 0.19   |
| BMW Group               | 1,996          | 6.35   | 793            | 4.54   | 4,208           | 7.94   | 6,997        | 6.87   |
| Chery Automobile        | 50             | 0.16   | 8              | 0.05   | 50              | 0.09   | 108          | 0.11   |
| Daimler Group           | 1,527          | 4.87   | 546            | 3.12   | 4,599           | 8.67   | 6,672        | 6.55   |
| FCA                     | 1,284          | 4.09   | 1,867          | 10.68  | 4,389           | 8.28   | 7,540        | 7.40   |
| Ford Group              | 1,373          | 4.38   | 760            | 4.35   | 2,558           | 4.82   | 4,691        | 4.61   |
| GM Group                | 2,426          | 7.74   | 1,376          | 7.87   | 2,975           | 5.61   | 6,777        | 6.65   |
| Geely Holding Group     | 1,148          | 3.66   | –              | –      | 1,734           | 3.27   | 2,882        | 2.83   |
| GWM                     | –              | –      | –              | –      | 50              | 0.09   | 50           | 0.05   |
| Honda                   | 1,284          | 4.09   | 451            | 2.85   | 1,071           | 2.02   | 2,806        | 2.75   |
| Hyundai Kia Group       | 2,677          | 8.54   | 1,587          | 9.08   | 3,282           | 6.19   | 7,546        | 7.41   |
| Isuzu                   | –              | –      | –              | –      | 22              | 0.04   | 22           | 0.02   |
| Mahindra & Mahindra     | –              | –      | –              | –      | 578             | 1.09   | 578          | 0.57   |
| Mazda                   | 1,245          | 3.97   | 419            | 2.40   | 947             | 1.79   | 2,611        | 2.56   |
| Mitsubishi              | 356            | 1.14   | 578            | 3.31   | 1,273           | 2.40   | 2,207        | 2.17   |
| PSA                     | 2,396          | 7.64   | 2,310          | 13.22  | 2,123           | 4.00   | 6,829        | 6.70   |
| Proton                  | 242            | 0.77   | –              | –      | 24              | 0.05   | 266          | 0.26   |
| Renault-Nissan Alliance | 2,389          | 7.62   | 2,126          | 12.17  | 5,219           | 9.84   | 9,734        | 9.56   |
| SAIC                    | 25             | 0.08   | 18             | 0.10   | 74              | 0.14   | 43           | 0.04   |
| Small and Medium OEM    | 6              | 0.02   | –              | –      | 1,036           | 1.31   | 80           | 0.08   |
| Subaru                  | 1,063          | 3.39   | 50             | 0.27   | 1,022           | 1.93   | 2,085        | 2.05   |
| Suzuki                  | 330            | 1.05   | 894            | 5.12   | 854             | 1.61   | 2,078        | 2.04   |
| Tata Group              | 103            | 0.33   | 32             | 0.18   | 3,534           | 6.67   | 3,669        | 3.60   |
| Tesla                   | –              | –      | –              | –      | 224             | 0.42   | 224          | 0.22   |
| Toyota Group            | 3,059          | 9.75   | 1,065          | 6.09   | 3,584           | 6.76   | 7,708        | 7.57   |
| VW Group                | 6,380          | 20.35  | 2,645          | 15.14  | 8,431           | 15.90  | 17,456       | 17.14  |
| All                     | 31,359         | 100.00 | 17,475         | 100.00 | 53,018          | 100.00 | 101,852      | 100.00 |

*Note:* Car segments follow the European passenger car classification: Segments A and B: *mini* and *small* cars; Segments C and D: *medium* and *large* cars; *Higher Segments* is an aggregate of segments E, F, MPV, and SUV.

Table A.3

Definitions of treatment groups.

|                 | Treatment group 1 ( $g=1$ )<br><b>Definition:</b> VWG Models with Defeat Device             | Treatment group 2 ( $g=2$ )<br><b>Definition:</b> All VW Models                                                  | Treatment group 3 ( $g=3$ )<br><b>Definition:</b> All VWG models                                                                                                                                                                                                                                                                                            |
|-----------------|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Segments A & B  |                                                                                             | Lupo, Fox, Polo, Up!                                                                                             | Audi A1, Audi S1, Citigo, Fabia Ibiza, Lupo, Mii, Fox, Polo, Up!                                                                                                                                                                                                                                                                                            |
| Segments C & D  | Audi A3, Beetle, Beetle Convertible, Golf, Golf Sportwagen, Jetta, Jetta Sportwagen, Passat | Beetle, Beetle Convertible, Eos, Golf, Golf Sportwagen, Jetta, Jetta Sportwagen, Passat, Santana Scirocco, Vento | Audi A3, Audi A4, Audi A5, Audi RS4, Audi RS5, Audi S3, Audi S4, Audi S5, Beetle, Eos, Exeo, Golf, Golf Sportwagen, Jetta, Jetta Sportwagen, Laura, Leon, Octavia, Passat, Santana, Superb, Rapid, Scirocco, Toledo, TT, Vento                                                                                                                              |
| Higher segments | Audi A6, Audi A7, Audi A8, Audi Q5, Audi Q7, Touareg, Cayenne                               | Phaeton, Saverio, Sharan, Tiguan, Touareg, Touran                                                                | Alhambra, Altea, Amarok, Audi A6, Audi A7, Audi A8, Audi Q3, Audi Q5, Audi Q7, Audi R8, Audi RS6, Audi RS7, Audi S6, Audi S7, Audi S8, Audi SQ5, Aventador, Bentley Continental, Boxster, Cayenne, Cayman, Continental, Flying Spur, Huracan, Macan, Mulsanne, Panamera, Phaeton, Porsche 911, Porsche 918, Roomster, Sharan, Tiguan, Touareg, Touran, Yeti |

Table A.4

Drivers – Environmental concern measure [Franzen and Vogl \(2013\)](#) – Segments C & D.

|                            | Segments C & D                              |                               |                      |                        |                       |
|----------------------------|---------------------------------------------|-------------------------------|----------------------|------------------------|-----------------------|
|                            | Treatment group 1                           |                               | Treatment group 2    |                        | Treatment group 3     |
|                            | <within-group>                              | <across-groups>               | <within-group>       | <across-groups>        | <across-groups>       |
|                            | (i)                                         | (ii)                          | (iii)                | (iv)                   | (v)                   |
| <i>Treated Units</i>       | VWG Models with Defeat Device               | VWG Models with Defeat Device | All VW Models        | All VW Models          | All VWG Models        |
| <i>TreatXPost</i>          | −0.1994<br>(0.1350)                         | 0.0770<br>(0.1274)            | −0.2476*<br>(0.1351) | 0.0836<br>(0.1366)     | 0.2341<br>(0.1364)    |
| <i>TreatXPostXEnvConc</i>  | −0.4055**<br>(0.2002)                       | −0.4236***<br>(0.0904)        | −0.3166<br>(0.1977)  | −0.3040***<br>(0.0957) | −0.1479<br>(0.0949)   |
| <i>TreatXPostXCulTight</i> | 0.3267*<br>(0.1966)                         | −0.0047<br>(0.1262)           | 0.2708<br>(0.1974)   | −0.1019<br>(0.1314)    | −0.2510**<br>(0.1317) |
| <i>Control Units</i>       | Other VWG Models                            | Non-VWG Models                | Other VWG Models     | Non-VWG Models         | Non-VWG Models        |
| <i>Observations</i>        | 3,906                                       | 19,930                        | 3,906                | 18,071                 | 20,329                |
| <i>R-squared</i>           | 0.2922                                      | 0.4149                        | 0.2928               | 0.4355                 | 0.4166                |
|                            | Segments C & D - Excluding “Alleged” Groups |                               |                      |                        |                       |
|                            | Treatment group 1                           |                               | Treatment group 2    |                        | Treatment group 3     |
|                            | <within-group>                              | <across-groups>               | <within-group>       | <across-groups>        | <across-groups>       |
|                            | (i)                                         | (ii)                          | (iii)                | (iv)                   | (v)                   |
| <i>Treated Units</i>       | VWG Models with Defeat Device               | VWG Models with Defeat Device | All VW Models        | All VW Models          | All VWG Models        |
| <i>TreatXPost</i>          | −0.1994<br>(0.1350)                         | −0.0634<br>(0.0935)           | −0.2476*<br>(0.1351) | −0.0658<br>(0.1066)    | 0.0853<br>(0.1066)    |
| <i>TreatXPostXEnvConc</i>  | −0.4055**<br>(0.2002)                       | −0.3839***<br>(0.1213)        | −0.3166<br>(0.1977)  | −0.2393*<br>(0.1236)   | −0.0834<br>(0.1230)   |
| <i>TreatXPostXCulTight</i> | 0.3267*<br>(0.1966)                         | 0.0429<br>(0.1061)            | 0.2708<br>(0.1974)   | −0.0746<br>(0.1019)    | −0.2244**<br>(0.1017) |
| <i>Control Units</i>       | Other VWG Models                            | Non-VWG Models                | Other VWG Models     | Non-VWG Models         | Non-VWG Models        |
| <i>Observations</i>        | 3,906                                       | 15,357                        | 3,906                | 13,498                 | 15,756                |
| <i>R-squared</i>           | 0.2922                                      | 0.4094                        | 0.2928               | 0.4352                 | 0.4104                |

Note: The dependent variable is the change in the monthly sales volumes of a car-model registered in a country. This change is measured as the first difference in the logarithm of sales volumes (on the corresponding month of the previous year). The variable “*EnvConc*” is the *environmental concern index* from [Franzen and Vogl \(2013\)](#) available for the following sub-sample of countries: Australia, Austria, Canada, Finland, Israel, Mexico, Slovakia, Spain, Sweden, Turkey, United Kingdom, United States. The variable “*CulTight*” is the *cultural tightness index* measuring the *pervasiveness of social norms* in a country on a unit-scale. Standard errors are robust and clustered at the group level in columns (ii), (iv) and (v). All specifications include a post-treatment indicator and its interactions with *EnvConc* and *CulTight*, alongside country-month, country-month, and month-model fixed effects. Segments C and D (*medium and large cars*) are segment classes of the European passenger car classification. The set of “Alleged” Groups includes BMW Group, Daimler Group, FCA, Renault-Nissan Alliance. \*, \*\*, and \*\*\* denote significance at the 10, 5, and 1 percentage level, respectively.

**Table A.5**

Scandal effect – Segments C &amp; D – Excluding Israel.

| Segments C & D (Sep 2014 - Sep 2016)                              |                                     |                                     |                        |                      |                     |
|-------------------------------------------------------------------|-------------------------------------|-------------------------------------|------------------------|----------------------|---------------------|
|                                                                   | Treatment group 1                   |                                     | Treatment group 2      |                      | Treatment group 3   |
|                                                                   | <within-group>                      | <across-group>                      | <within-group>         | <across-groups>      | <across-groups>     |
|                                                                   | (i)                                 | (ii)                                | (iii)                  | (iv)                 | (v)                 |
| <i>Treated Units</i>                                              | VWG<br>Models with<br>Defeat Device | VWG<br>Models with<br>Defeat Device | All VW<br>Models       | All VW<br>Models     | All VWG<br>Models   |
| <i>TreatXPost</i>                                                 | −0.1365***<br>(0.0457)              | −0.0506<br>(0.0720)                 | −0.1421***<br>(0.0454) | −0.0291<br>(0.0787)  | 0.0589<br>(0.0786)  |
| <i>Control Units</i>                                              | Other VWG<br>Models                 | Non-VWG<br>Models                   | Other VWG<br>Models    | Non-VWG<br>Models    | Non-VWG<br>Models   |
| <i>Observations</i>                                               | 5,977                               | 28,961                              | 5,977                  | 25,929               | 29,540              |
| <i>R-squared</i>                                                  | 0.2906                              | 0.4250                              | 0.2907                 | 0.4434               | 0.4242              |
| Segments C & D - Excluding “Alleged” Groups (Sep 2014 - Sep 2016) |                                     |                                     |                        |                      |                     |
|                                                                   | Treatment group 1                   |                                     | Treatment group 2      |                      | Treatment group 3   |
|                                                                   | <within-group>                      | <across-group>                      | <within-group>         | <across-groups>      | <across-groups>     |
|                                                                   | (i)                                 | (ii)                                | (iii)                  | (iv)                 | (v)                 |
| <i>Treated Units</i>                                              | VWG<br>Models with<br>Defeat Device | VWG<br>Models with<br>Defeat Device | All VW<br>Models       | All VW<br>Models     | All VWG<br>Models   |
| <i>TreatXPost</i>                                                 | −0.1365***<br>(0.0457)              | −0.1432**<br>(0.0546)               | −0.1421***<br>(0.0454) | −0.1305*<br>(0.0643) | −0.0427<br>(0.0641) |
| <i>Control Units</i>                                              | Other VWG<br>Models                 | Non-VWG<br>Models                   | Other VWG<br>Models    | Non-VWG<br>Models    | Non-VWG<br>Models   |
| <i>Observations</i>                                               | 5,977                               | 22,165                              | 5,977                  | 19,133               | 22,744              |
| <i>R-squared</i>                                                  | 0.2906                              | 0.4107                              | 0.2907                 | 0.4344               | 0.4091              |

Note: The estimation sample excludes observations from Israel (sensitivity analysis). The dependent variable is the change in the monthly sales volumes of a car-model registered in a country. This change is measured as the first difference in the logarithm of sales volumes (on the corresponding month of the previous year). Standard errors are robust and clustered at the group level in columns (ii), (iv) and (v). All specifications include a post-treatment indicator, along with country-month, country-model and month-model fixed effects. Segments C and D (*medium* and *large* cars) are segment classes of the European passenger car classification. The set of “Alleged” Groups includes BMW Group, Daimler Group, FCA, Renault-Nissan Alliance. \*, \*\*, and \*\*\* denote significance at the 10, 5, and 1 percentage level, respectively.

**Table A.6**

Drivers - Environmental concern vs cultural tightness - Segments C &amp; D - Excluding Israel.

| Segments C & D             |                                     |                                     |                        |                        |                       |
|----------------------------|-------------------------------------|-------------------------------------|------------------------|------------------------|-----------------------|
|                            | Treatment group 1                   |                                     | Treatment group 2      |                        | Treatment group 3     |
|                            | <within-group>                      | <across-groups>                     | <within-group>         | <across-groups>        | <across-groups>       |
|                            | (i)                                 | (ii)                                | (iii)                  | (iv)                   | (v)                   |
| <i>Treated Units</i>       | VWG<br>Models with<br>Defeat Device | VWG<br>Models with<br>Defeat Device | All VW<br>Models       | All VW<br>Models       | All VWG<br>Models     |
| <i>TreatXPost</i>          | −0.0202<br>(0.1052)                 | 0.1723<br>(0.1043)                  | 0.1074<br>(0.1064)     | 0.2826**<br>(0.1173)   | 0.2114*<br>(0.1173)   |
| <i>TreatXPostXEnvConc</i>  | −0.4790***<br>(0.1566)              | −0.4850***<br>(0.0832)              | −0.6178***<br>(0.1544) | −0.5160***<br>(0.0900) | −0.1743*<br>(0.0898)  |
| <i>TreatXPostXCulTight</i> | 0.1650<br>(0.1573)                  | −0.0684<br>(0.0824)                 | 0.0041<br>(0.1590)     | −0.2240**<br>(0.0870)  | −0.1939**<br>(0.0879) |
| <i>Control Units</i>       | Other VWG<br>Models                 | Non-VWG<br>Models                   | Other VWG<br>Models    | Non-VWG<br>Models      | Non-VWG<br>Models     |
| <i>Observations</i>        | 5,977                               | 28,961                              | 5,977                  | 25,929                 | 29,540                |
| <i>R-squared</i>           | 0.2943                              | 0.4256                              | 0.2950                 | 0.4442                 | 0.4247                |

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Table A.6 (continued).

|                            | Segments C & D - Excluding "Alleged" Groups |                        |                        |                        |                       |
|----------------------------|---------------------------------------------|------------------------|------------------------|------------------------|-----------------------|
|                            | Treatment group 1                           |                        | Treatment group 2      |                        | Treatment group 3     |
|                            | <within-group>                              | <across-groups>        | <within-group>         | <across-groups>        | <across-groups>       |
|                            | (i)                                         | (ii)                   | (iii)                  | (iv)                   | (v)                   |
| <i>TreatXPost</i>          | −0.0202<br>(0.1052)                         | 0.0471<br>(0.0893)     | 0.1074<br>(0.1064)     | 0.1411<br>(0.1092)     | 0.0695<br>(0.1089)    |
| <i>TreatXPostXEnvConc</i>  | −0.4790***<br>(0.1566)                      | −0.4165***<br>(0.1075) | −0.6178***<br>(0.1544) | −0.4165***<br>(0.1077) | −0.0745<br>(0.1074)   |
| <i>TreatXPostXCulTight</i> | 0.1650<br>(0.1573)                          | −0.0587<br>(0.0714)    | 0.0041<br>(0.1590)     | −0.2286***<br>(0.0709) | −0.1982**<br>(0.0713) |
| <i>Control Units</i>       | Other VWG<br>Models                         | Non-VWG<br>Models      | Other VWG<br>Models    | Non-VWG<br>Models      | Non-VWG<br>Models     |
| <i>Observations</i>        | 5,977                                       | 22,165                 | 5,977                  | 19,133                 | 22,744                |
| <i>R-squared</i>           | 0.2943                                      | 0.4116                 | 0.2950                 | 0.4358                 | 0.4100                |

Note: The estimation sample excludes observations from Israel (sensitivity analysis). The dependent variable is the change in the monthly sales volumes of a car-model registered in a country. This change is measured as the first difference in the logarithm of sales volumes (on the corresponding month of the previous year). The variable "*EnvConc*" is the *environmental concern index* expressing customers' "environmental concern" in a unit-scale. The variable "*CulTight*" is the *cultural tightness index* measuring the *pervasiveness of social norms* in a country on a unit-scale. Standard errors are robust and clustered at the group level in columns (ii), (iv) and (v). All specifications include a post-treatment indicator and its interactions with *EnvConc* and *CulTight*, alongside country-month, country-model, and month-model fixed effects. Segments C and D (*medium* and *large* cars) are segment classes of the European passenger car classification. The set of "Alleged" Groups includes BMW Group, Daimler Group, FCA, Renault-Nissan Alliance. \*, \*\*, and \*\*\* denote significance at the 10, 5, and 1 percentage level, respectively.

Table A.7

Drivers - Environmental concern vs cultural tightness - Segments C & D - Sales Bans.

|                            | Segments C & D                              |                                     |                        |                        |                        |
|----------------------------|---------------------------------------------|-------------------------------------|------------------------|------------------------|------------------------|
|                            | Treatment group 1                           |                                     | Treatment group 2      |                        | Treatment group 3      |
|                            | <within-group>                              | <across-groups>                     | <within-group>         | <across-groups>        | <across-groups>        |
|                            | (i)                                         | (ii)                                | (iii)                  | (iv)                   | (v)                    |
| <i>Treated Units</i>       | VWG<br>Models with<br>Defeat Device         | VWG<br>Models with<br>Defeat Device | All VW<br>Models       | All VW<br>Models       | All VWG<br>Models      |
| <i>TreatXPost</i>          | 0.1693<br>(0.1082)                          | 0.3608**<br>(0.1376)                | 0.0988<br>(0.1083)     | 0.3415**<br>(0.1557)   | 0.2791*<br>(0.1557)    |
| <i>TreatXPostXEnvConc</i>  | −0.5494***<br>(0.1664)                      | −0.7062***<br>(0.1123)              | −0.5292***<br>(0.1653) | −0.6394***<br>(0.1279) | −0.3783***<br>(0.1277) |
| <i>TreatXPostXCulTight</i> | 0.0023<br>(0.1574)                          | −0.2106*<br>(0.1186)                | −0.0356<br>(0.1583)    | −0.2807**<br>(0.1309)  | −0.2235<br>(0.1320)    |
| <i>Control Units</i>       | Other VWG<br>Models                         | Non-VWG<br>Models                   | Other VWG<br>Models    | Non-VWG<br>Models      | Non-VWG<br>Models      |
| <i>Observations</i>        | 4,589                                       | 21,998                              | 4,589                  | 19,698                 | 22,462                 |
| <i>R-squared</i>           | 0.2787                                      | 0.3744                              | 0.2794                 | 0.3883                 | 0.3755                 |
|                            | Segments C & D - Excluding "Alleged" Groups |                                     |                        |                        |                        |
|                            | Treatment group 1                           |                                     | Treatment group 2      |                        | Treatment group 3      |
|                            | <within-group>                              | <across-groups>                     | <within-group>         | <across-groups>        | <across-groups>        |
|                            | (i)                                         | (ii)                                | (iii)                  | (iv)                   | (v)                    |
| <i>TreatXPost</i>          | 0.1693<br>(0.1082)                          | 0.2401*<br>(0.1247)                 | 0.0988<br>(0.1083)     | 0.2059<br>(0.1511)     | 0.1449<br>(0.1513)     |
| <i>TreatXPostXEnvConc</i>  | −0.5494***<br>(0.1664)                      | −0.6024***<br>(0.1325)              | −0.5292***<br>(0.1653) | −0.5058***<br>(0.1479) | −0.2462<br>(0.1483)    |
| <i>TreatXPostXCulTight</i> | 0.0023<br>(0.1574)                          | −0.2049*<br>(0.1031)                | −0.0356<br>(0.1583)    | −0.2896**<br>(0.1160)  | −0.2341*<br>(0.1166)   |

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Table A.7 (continued).

| Control Units | Other VWG Models | Non-VWG Models | Other VWG Models | Non-VWG Models | Non-VWG Models |
|---------------|------------------|----------------|------------------|----------------|----------------|
| Observations  | 4,589            | 16,877         | 4,589            | 14,577         | 17,341         |
| R-squared     | 0.2787           | 0.3547         | 0.2794           | 0.3720         | 0.3565         |

Note: The estimation sample excludes countries that have implemented sales bans (or similar interventions) on cars equipped with defeat devices after the scandal (sensitivity analysis). The dependent variable is the change in the monthly sales volumes of a car-model registered in a country. This change is measured as the first difference in the logarithm of sales volumes (on the corresponding month of the previous year). The variable “*EnvConc*” is the *environmental concern index* expressing customers’ “environmental concern” in a unit-scale. The variable “*CulTight*” is the *cultural tightness index* measuring the *pervasiveness of social norms* in a country on a unit-scale. Standard errors are robust and clustered at the group level in columns (ii), (iv) and (v). All specifications include a post-treatment indicator and its interactions with *EnvConc* and *CulTight*, alongside country-month, country-model, and month-model fixed effects. Segments C and D (*medium* and *large* cars) are segment classes of the European passenger car classification. The set of “Alleged” Groups includes BMW Group, Daimler Group, FCA, Renault-Nissan Alliance. \*, \*\*, and \*\*\* denote significance at the 10, 5, and 1 percentage level, respectively.

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